

Electrical Circuits Laboratory - Circuit Questions & Solutions

Clean answer sheet with choices and worked solutions. The source watermark/logo is not reproduced.

1) Nodal voltage V_o (10 V source, 10 Ω , 15 Ω and 2 A source)

The circuit has a 10 V node connected to V_o through 10 Ω , V_o connected to ground through 15 Ω , and a 2 A current source directed from the right node toward V_o .

Choices	Option
A	10 V
B	6 V
C	18 V
D	12 V

Solution: Apply KCL at V_o : $(V_o - 10)/10 + V_o/15 - 2 = 0$. Multiplying by 30 gives $3V_o - 30 + 2V_o - 60 = 0$, hence $5V_o = 90$ and $V_o = 18$ V.

Correct answer: C — 18 V

2) Equivalent resistance seen by the source (40 Ω network)

The 40 Ω branch is in parallel with a second branch containing 6 Ω in series with (12 Ω || 6 Ω). The result is then in series with the external 10 Ω and 22 Ω resistors.

Choices	Option
A	40 Ω
B	18 Ω
C	8 Ω
D	26 Ω

Solution: First, $12 \parallel 6 = (12 \times 6)/(12 + 6) = 4$ Ω . The lower branch is $6 + 4 = 10$ Ω . Therefore, $40 \parallel 10 = (40 \times 10)/(40 + 10) = 8$ Ω . Total resistance = $10 + 8 + 22 = 40$ Ω .

Correct answer: A — 40 Ω

3) Equivalent resistance seen by the source (12 Ω , 2 Ω , 8 Ω , 2.4 Ω and 9 Ω)

The 2 Ω and 8 Ω resistors are parallel. Their equivalent is in series with 2.4 Ω . That branch is parallel with 12 Ω , and the resulting network is in series with 9 Ω .

Choices	Option
A	9 Ω
B	15 Ω

C	12 Ω
D	3 Ω

Solution: $2 \parallel 8 = (2 \times 8)/(2+8) = 1.6 \Omega$. Then $1.6 + 2.4 = 4 \Omega$. Next, $12 \parallel 4 = (12 \times 4)/(12+4) = 3 \Omega$. Finally, $9 + 3 = 12 \Omega$.

Correct answer: C — 12 Ω

4) Current i due to the 1.2 A current source only (superposition)

Deactivate the independent voltage source by replacing it with a short circuit. The 6 Ω , 2 Ω and 3 Ω resistors are then connected in parallel with the 1.2 A source. Current i is the current through the 2 Ω resistor.

Choices	Option
A	2.4 A
B	1.2 A
C	0.6 A
D	1 A

Solution: Conductance: $1/6 + 1/2 + 1/3 = 1 \text{ S}$. Node voltage = $I/G = 1.2/1 = 1.2 \text{ V}$. Therefore, $i = V/2 = 1.2/2 = 0.6 \text{ A}$.

Correct answer: C — 0.6 A

5) KCL relation at the node

At the node, i_1 and i_2 enter the node, while i_3 , i_4 and i_5 leave the node.

Choices	Option
A	$i_1 + i_4 + i_5 = i_2 + i_3$
B	$i_1 + i_2 = i_3 + i_4 + i_5$
C	$i_1 + i_2 + i_3 = i_4 + i_5$
D	$i_1 = i_2 + i_3 + i_4 + i_5$

Solution: Kirchhoff's Current Law states that the sum of currents entering a node equals the sum of currents leaving it. Thus, $i_1 + i_2 = i_3 + i_4 + i_5$.

Correct answer: B — $i_1 + i_2 = i_3 + i_4 + i_5$

Important circuit formulas used

Series resistors	$R_{eq} = R_1 + R_2 + \dots$
Parallel resistors	$R_{eq} = (R_1 R_2) / (R_1 + R_2)$ for two resistors
Ohm's law	$V = IR$
KCL	$\Sigma I_{entering} = \Sigma I_{leaving}$
Nodal analysis	Sum of currents at the node = 0 (with signs)
Superposition	Deactivate voltage sources $\rightarrow$ short circuit; current sources $\rightarrow$ open circuit

Note: The solutions above are based on the visible circuit directions and component values in the supplied pages. If a scanned arrow/polarity differs, the corresponding sign may change.